

MATERIALS AND METHODOLOGY

INTRODUCTION

In this chapter, the methods and procedure adopted to carry out the experiments have been discussed. Details about materials, Equipments used and tests conducted have been given. The chapter consists of materials and experimental procedure.

MATERIALS AND EQUIPMENTS USED

A 350 GSM plain woven bleached cotton fabric of 44 EPI (ends/ inch) and 22 PPI (picks/inch) with 10/2s warp and 8/2s weft count was used in this study.

Spinach juice (SJ)

Coconut shell extract (CSE) were extracted from fresh spinach leaves and waste coconut shells. Extracted juices were filtered using Whatman No 1 filter paper and prepared for application on cotton fabric. Anhydrous sodium carbonate (Na_2CO_3) was used for the treatment.

Original fresh CSE solution after extraction looked yellowish brown in color and showed the pH value of 4.5. It was made alkaline (pH 10) by the addition of anhydrous sodium carbonate (Na_2CO_3).

TEST CONDUCTED:

Extraction of Spinach Juice

Spinach Juice used for study was extracted from fresh spinach leaves then filtration was carried out for further application. spinach leaf solution after extraction looked green in color and had a pH value of 5.5. It was made alkaline (pH 10) by the addition of anhydrous Na_2CO_3 .

Application on cotton fabric

Bleached fabrics were then impregnated with CSE juice, spinach juice (SJ) and mixture of CSE+SJ (1:1) at 90°C for 60 minutes with a material to liquor ratio of 1:15. After that, the treated fabric samples were air dried at room temperature. Later samples were conditioned for at least 36 hours at 65 percent relative humidity (RH) and 27 degrees Celsius prior to testing.

TESTING

3 Different samples is being tested, which is treated fabric and other is untreated (bleached fabric).

Testing is done using Digital Flammability Testing Machine.

METHODOLOGY

Step 1: Bleached fabric is being desized and scoured.

Desizing Recipe:-

HCl -1% on the weight of the material.

Wetting agent - 0.5%-1% on the weight of material

M:L:: 1:40

Time:-1hr

Temperature:- At room temperature

Scouring Recipe:-

NaOH-4%(CausticSoda)

Na₂CO₃-1%(SodaAsh)

WettingAgent-0.5 1gm/ltr

M:L::1:40

Temperature:-30°C

Time:-1hr

Step 2: Extraction of Coconut Shell Juice

Coconut shells are cleaned. And inner side of the shell is being extracted. And made it in the form of juice by adding , Water and sodium carbonate to it. In order to form the char formation.



Green Coconuts



Coconut Shell



Coconut Shell Extract (CSE)

Step 3: Extraction of Spinach Juice

Spinach is being boiled at room temperature for 30 mins until it gets soft by mixing water, then it is being blended and juice is being extracted.

After, the preparation of the juice ,it is being filtered and sodium carbonate is being added to it in order to get alkaline.



Spinach Leaves



Extracted Spinach Juice

Extraction of Juice and Filtration



Samples Of Flame Retardant Finish applied Fabrics



RESULT OF THE TEST

By the above result we can say that, as char formation increases, the flammability decreases By the above result we can say that, as char formation increases, the flammability decreases.

BURNING OF UNTREATED (BLEACHED FABRIC) IN SEC	COCONUT SHELL JUICE	SPINACH JUICE	CSE+ SI
10.45 Sec	12.54 Sec	14.75 Sec	13.89 Sec
10.54 Sec	13.15 Sec	12.89 Sec	13.94 Sec
10.12 Sec	13.32 Sec	13.50 Sec	13.97 Sec
10.16 Sec	14.02 Sec	15.06 Sec	14.05 Sec
10.20 Sec	14.56 Sec	15.57 Sec	14.55 Sec